EX 2:FORMAL DESIGN OF AN EMBEDDED CONTROLLER USING STATE MACHINE (STATECHART) MODELING – TRAFFIC LIGHT CONTROLLER.

CODE:

```
int red = 8;
int yellow = 9;
int green = 10;

void setup()
{
  pinMode(red, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(green, OUTPUT);
}

const int red = 8;
const int yellow = 9;
const int green = 10;

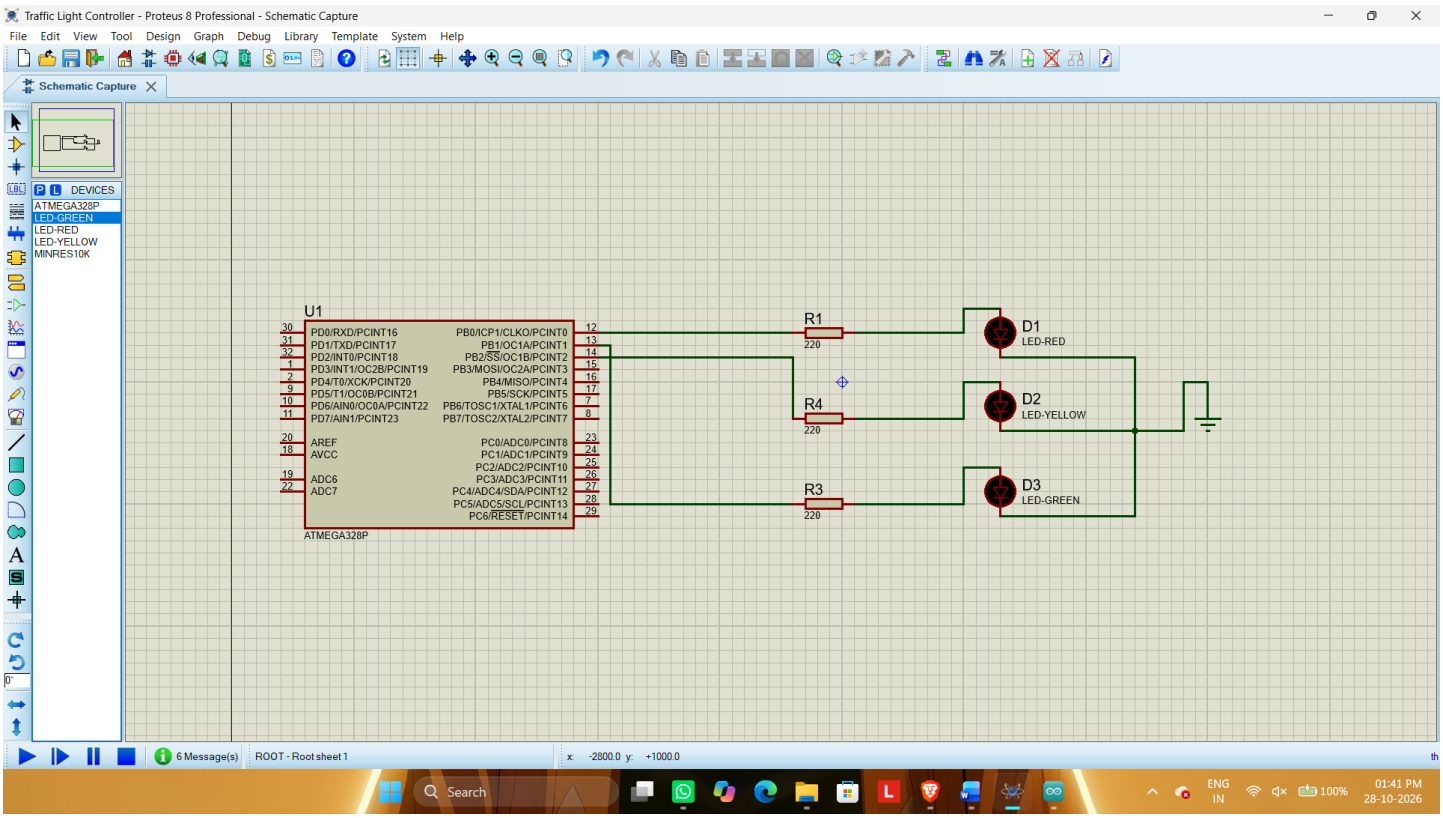
void setup() {
  pinMode(red, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(green, OUTPUT);

  digitalWrite(red, LOW);
  digitalWrite(yellow, LOW);
  digitalWrite(green, LOW);
}

void loop() {
  // RED ON
  digitalWrite(red, HIGH);
  digitalWrite(yellow, LOW);
  digitalWrite(green, LOW);
  delay(5000);
```

```
  // GREEN ON
  digitalWrite(red, LOW);
  digitalWrite(yellow, LOW);
  digitalWrite(green, HIGH);
  delay(5000);

  // YELLOW ON
  digitalWrite(red, LOW);
  digitalWrite(yellow, HIGH);
  digitalWrite(green, LOW);
  delay(2000);
}
```



Reflection

Through this project, I gained practical knowledge of designing and simulating an embedded traffic light controller using Arduino IDE and Proteus 8. I learned how to implement a traffic light sequence using a finite state machine approach, where each state represents a specific traffic light condition, such as Red, Green, or Yellow. I also learned how to write Arduino code, configure the ATmega328P microcontroller, connect LEDs with current-limiting resistors, and load the compiled HEX file into Proteus for simulation.

During the simulation, I faced challenges in troubleshooting LED operation, checking microcontroller settings, and verifying the program execution. These difficulties helped me understand the importance of

correct pin mapping, clock configuration, and circuit connections. I also learned that systematic testing and debugging are essential for developing a reliable embedded system.

Overall, this project improved my understanding of state machine modeling, microcontroller programming, and circuit simulation. It also helped me develop problem-solving skills and understand how state-based control systems can be applied to real-world traffic management applications.